



Question Set

Question 1

Consider an equity option under Merton jump diffusion. Define the state variables (diffusion volatility, Poisson intensity, jump mean, jump volatility, and compensator) and derive the jump-compensated risk-neutral SDE. State the measure, filtration, and tradability assumptions.

$$\frac{dS_t}{S_{t-}} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Question 2

For an equity option under Merton jump diffusion, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in an equity option under Merton jump diffusion. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using jump intensity 0.7, mean jump -8 percent, jump volatility 18 percent, and one-year maturity, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

$$\frac{dS_t}{S_{t-}} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Question 5

For an equity option under Merton jump diffusion, compute and interpret diffusion delta versus unhedgeable jump exposure. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes short-dated smile and crash put quotes. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for an equity option under Merton jump diffusion. Explain how compound Poisson distribution and terminal payoff prevents economically impossible prices.

$$\frac{dS_t}{S_{t-}} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Question 8

Design a Monte Carlo or lattice estimator for an equity option under Merton jump diffusion. Specify diffusion paths with random jump counts and jump sizes, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements an equity option under Merton jump diffusion and gets a stable but wrong answer. Explain how forgetting the compensator and therefore shifting the forward price creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for an equity option under Merton jump diffusion. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of an equity option under Merton jump diffusion, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of jump intensity 0.7, mean jump -8 percent, jump volatility 18 percent, and one-year maturity. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for an equity option under Merton jump diffusion. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in an equity option under Merton jump diffusion. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for an equity option under Merton jump diffusion is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for short-dated smile and crash put quotes are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For an equity option under Merton jump diffusion, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for diffusion paths with random jump counts and jump sizes. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that forgetting the compensator and therefore shifting the forward price could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

$$\frac{dS_t}{S_t} = (r - \lambda\kappa)dt + \sigma dW_t + (J - 1)dN_t$$

Question 20

Explain an equity option under Merton jump diffusion and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.